



Smart Device Services Manager

HP Print Quality Diagnostics

User Guide

Document Revision: V1.2



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Document Overview

HP SDS Manager Print Quality Diagnostics (PQD) allows for remote, and/or user-interactive print quality monitoring.

The exact mechanism available for analysing print quality depends upon the device model:

- Some models include an inbuilt hardware Inline Scan Bar
- Some models support the optional addition of a hardware Inline Scan Bar Add-On Accessory
- Some models support the optional installation of a software 'Print Quality Diagnostics App' that can be accessed from the device's control panel.

The mechanism for triggering print quality analysis varies according to the hardware, firmware, and software available in the device.

Regardless of the analysis mechanism used, any print quality faults found will result in the generation of a new type of engineering advisory, "[PrintQualityDiagnostics](#)", which can be accessed via the existing SDS Manager engineering advisory system.



Prerequisites

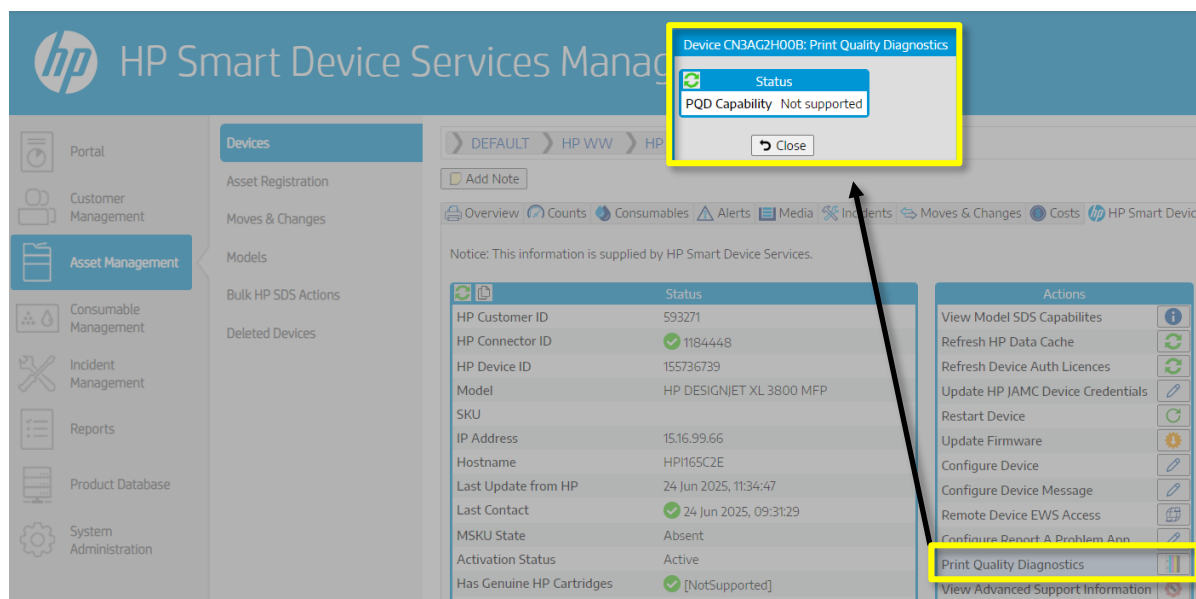
HP Print Quality Diagnostics is only available to a select number of device models - the full list can be obtained from the **HP Print Quality diagnostics Technical Detail** document (available for download from Fresh Desk).

Remote invocation of print quality diagnostics via Inline Scan Bar requires that on premises JAMC's be at version 4.1.6492 or above. If a device is being managed by a JAMC below that level, the operation will fail.

Remote installation of the print quality diagnostics app will require that on premises JAMC's be at or above version 4.1.6492. If a device is being managed by a JAMC below that level, the operation will fail.

Checking Device Capability

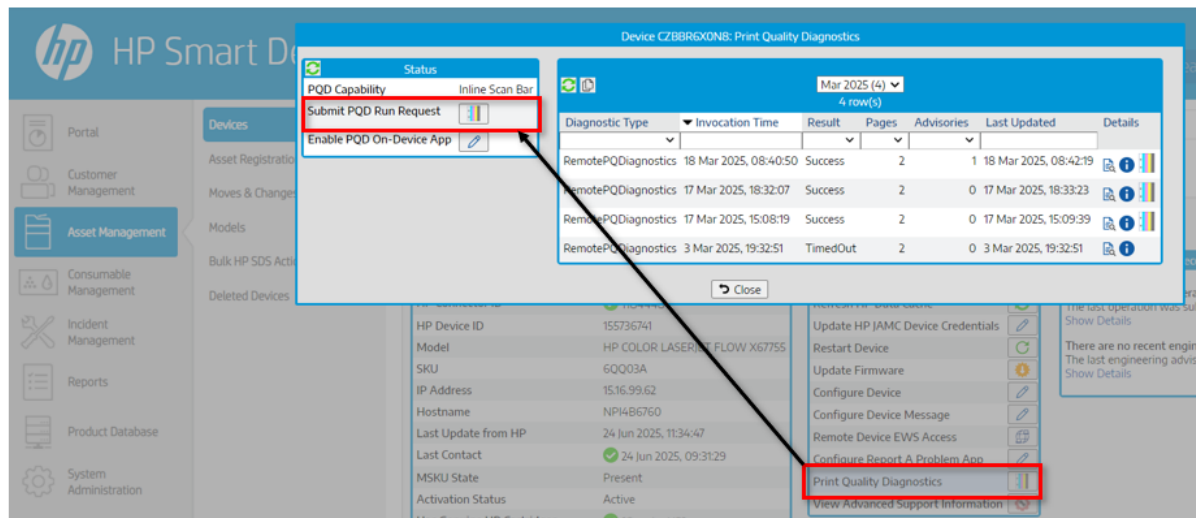
To check if your device is compatible, go to the SDS Manager portal device page, click on the "Print Quality Diagnostics" action button, and the PQD capability is displayed in the status box.



Triggering of Print Quality Analysis

Inline Scan Bar (Hardware) – Remote User Invoked Diagnostics

For devices with an Inline Scan Bar, either built-in or add-on accessory, print quality analysis can be remotely invoked by a manual action in the SDS Manager portal.



This will cause specially designed diagnostic test pages to be printed, scanned, and automatically uploaded to HP's cloud for analysis. *The **HP Print Quality Diagnostics Technical Details** document, available for download from Fresh Desk, describes the security features which protects against user documents from being accidentally uploaded to HP's cloud.*

Inline Scan Bar (Hardware) – Firmware Invoked Diagnostics

For devices with a **built-in** Inline Scan Bar, and the correct firmware version, the device firmware will automatically sample the non-printable leading edge of ordinary printed pages for local print quality issues. This process cannot be disabled, and the timing cannot be changed, but it only uploads print quality analysis results to HP's cloud. No scanned images or print content leaves the printer.

HP Print Quality Diagnostics App (Software) – Local User Invoked Diagnostics

The HP Print Quality Diagnostics app is located on the device control panel and print quality analysis can be performed manually from there.

Note: Users can install the HP Print Quality Diagnostics app onto their device remotely via the SDS Manager portal (see below) but print quality analysis can only be invoked from the app on the device control panel and not remotely from the portal.



Device CZBBR6X0N8: Print Quality Diagnostics

Status

PQD Capability Inline Scan Bar

Submit PQD Run Request

Enable PQD On-Device App

Jul 2025 (0)

0 row(s)

Diagnostic Type	Invocation Time	Result	Pages	Advisories	Last Updated	Details
-----------------	-----------------	--------	-------	------------	--------------	---------

Close

Notice: This information is supplied by HP Smart Device Services.

Status
HP Customer ID 623539
HP Connector ID 1454078
HP Device ID 156310505
Model HP COLOR LASERJET ELOW X67755
SKU 6QQ03A
IP Address 15.16.99.62
Hostname NPI4B6760
Last Update from HP 9 Jul 2025, 07:27:28
Last Contact 9 Jul 2025, 01:17:51
MSKU State Present
Activation Status Active

Actions

View Model SDS Capabilities

Refresh HP Data Cache

Update HP JAMC Device Credentials

Restart Device

Update Firmware

Configure Device

Configure Device Message

Remote Device EWS Access

Configure Report A Problem App

Print Quality Diagnostics

View Advanced Support Information

There are no re
The last operat
Show Details
There are no e

Use of the app involves a user manually scanning the specially designed diagnostic test pages which are printed as part of this process. These test pages are uploaded to HP's cloud for analysis. The **HP Print Quality Diagnostics Technical Details** document, available for download from Fresh Desk, describes the security features which protects against user documents from being accidentally uploaded to HP's cloud.

Perform print quality analysis from On Device Services on the device control panel

1. Tap on the **On Device Services** app.

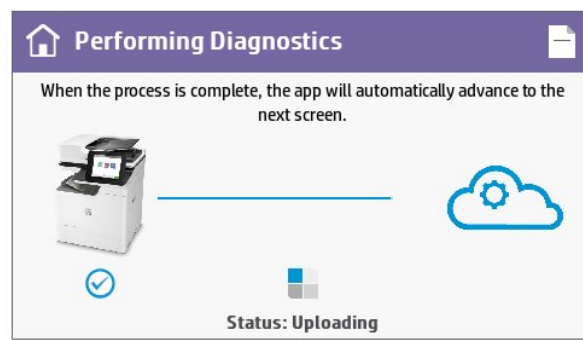




2. Tap on **HP Print Quality Diagnostics**.



3. Follow the on-screen instructions to print and scan a diagnostic test page.
4. The Performing Diagnostics window presents the status of the diagnostic run.



Viewing Audit Logs of Manually Invoked Analysis Runs

HP log all **manually** invoked diagnostics runs, including information about the time and mechanism of invocation, the number of test pages printed, the success of the run, plus triage information if for some reason the run failed. Also included are links to the scanned images of the test pages, and links to any Engineering Advisories that were generated if any print quality faults were found. These audit logs can be viewed from the SDS Manager portal.

1. Click the “Print Quality Diagnostics” action button to view a list of audit logs.



Device MXBCQ7B081: Print Quality Diagnostics

Status

PQD Capability Inline Scan Bar

Submit PQD Run Request

Disable PQD App

Jun 2024 (7)
 7 row(s)

Diagnostic Type	Invocation Time	Result	Pages	Advisories	Last Updated	Details
OnDeviceAdfDiagnostics	11 Jun 2024, 18:40:56	Success	2	0	11 Jun 2024, 18:41:47	
OnDeviceAdfDiagnostics	5 Jun 2024, 22:08:38	Success	2	0	5 Jun 2024, 22:09:41	
OnDeviceAdfDiagnostics	4 Jun 2024, 19:04:39	Success	2	0	4 Jun 2024, 19:05:45	
RemotePQDiagnostics	4 Jun 2024, 18:10:30	Success	2	0	4 Jun 2024, 18:11:58	
OnDeviceAdfDiagnostics	4 Jun 2024, 18:08:47	Success	2	1	4 Jun 2024, 18:09:22	
OnDeviceAdfDiagnostics	4 Jun 2024, 18:08:02	ImageAnalysisFailed	2	0	4 Jun 2024, 18:08:31	
OnDeviceAdfDiagnostics	4 Jun 2024, 18:06:47	ImageAnalysisFailed	2	0	4 Jun 2024, 18:07:19	

Close

IP Address: 154.65.17.1
Hostname: NPID97AC5.boi.rd.hpcorp.net
Last Update from HP: 8 Jul 2025, 09:18:29
Last Contact: 8 Jul 2025, 09:10:48
MSKU State: Present
Activation Status: Active

Configure Device
Configure Device Message
Remote Device EWS Access
Configure Report A Problem App
Print Quality Diagnostics
View Advanced Support Information

- Click on the “Details” button to view the details of a specific diagnostic run.

Device MXBCQ7B081: Print Quality Diagnostics

Audit Log Details

HP PQD Audit Log Id: 1C1E3B50BF6243F69A53B9430EC896FB

Diagnostic Type: OnDeviceAdfDiagnostics

Invocation Time: 4 Jun 2024, 18:08:47

Result: Success

Pages: 2

Engineering Advisories: **PQTriageLightPrintYellow**

Last Updated: 4 Jun 2024, 18:09:22

Printed Images: **Download**

Close

Links to the Engineering Advisories and the diagnostic images

NOTE: Audit Log entries are available for 18 months after creation and are then deleted. Operation Audit Log entries older than 18 months are not available.

NOTE: Automatic firmware invoked diagnostics does not involve the uploading of test pages and is excluded from audit logs.



Viewing and Downloading Scanned Images from Analysis Runs

Device CNBRR683PR: Print Quality Diagnostics

Status: PQR Capability App enabled, Disable PQR App

Jun 2025 (2) 2 row(s)

Diagnostic Type	Invocation Time	Result	Pages	Advisories	Last Updated	Details
OnDeviceAdfDiagnostics	04-Jun-2025 06:12:41	Success	2	0	04-Jun-2025 06:13:47	[Document] [Info] [Print Quality]
OnDeviceAdfDiagnostics	04-Jun-2025 01:27:52	Success	2	1	04-Jun-2025 01:29:46	[Document] [Info] [Print Quality]

Close

Only HP specific print quality test pages printed as part of manual diagnostic runs are available to view and download, and they can be obtained from the “Details” column of audit logs.

To view the images directly from the portal click on the print quality button from the “Details” column.

Device CNBRR683PR: Print Quality Diagnostics

Status: PQR Capability App enabled, Disable PQR App

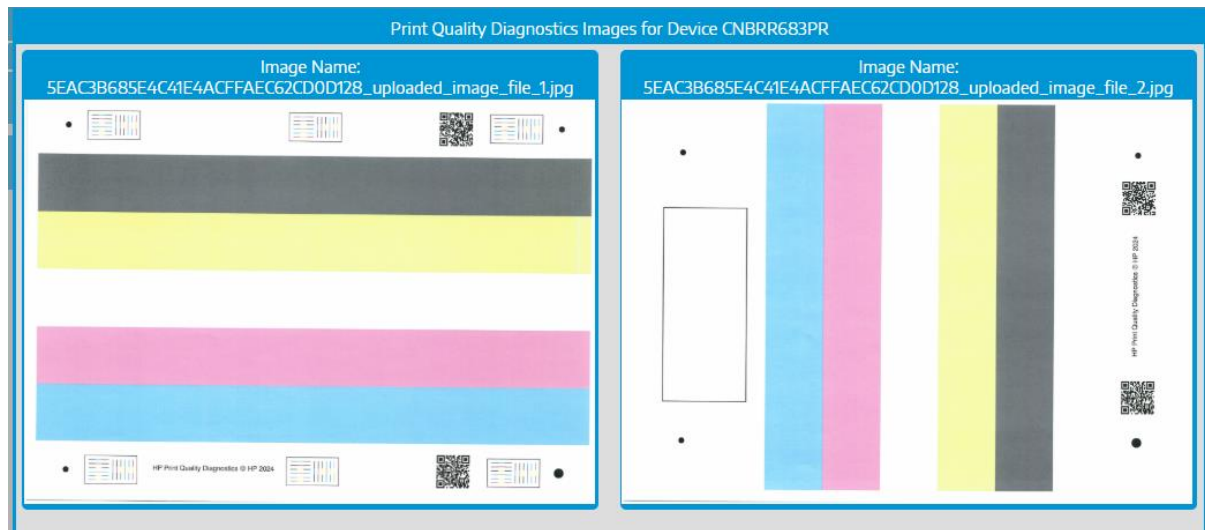
Jun 2025 (2) 2 row(s)

Diagnostic Type	Invocation Time	Result	Pages	Advisories	Last Updated	Details
OnDeviceAdfDiagnostics	04-Jun-2025 06:12:41	Success	2	0	04-Jun-2025 06:13:47	[Document] [Info] [Print Quality]
OnDeviceAdfDiagnostics	04-Jun-2025 01:27:52	Success	2	1	04-Jun-2025 01:29:46	[Document] [Info] [Print Quality]

Close

Status: HP Customer ID 513771, HP Connector ID 1463405, HP Device ID 156252241, Model HP COLOR LASERJET MFP E78625, SKU 5QJ90A, IP Address 192.168.8.167, Hostname wpp.localhost.12350.AQAAAAGOmGrgAgAADa3LOfL1, Last Update from HP 09-Jul-2025 15:07:50, Last Contact 09-Jul-2025 10:12:11, Ping Last Contact 09-Jul-2025 15:27:48

Actions: View Model SDS Capabilities, Refresh HP Data Cache, Restart Device, Update Firmware, Configure Device, Configure Device Message, Remote Device EWS Access, Configure Report A Problem App, **Print Quality Diagnostics**, View Advanced Support Information



Higher resolution versions of the images are available for download and viewing locally by using the “Download” option from the “Audit Log Details”.

Device CNBRR683PR: Print Quality Diagnostics

Status

PQD Capability App enabled

Disable PQD App

Jun 2025 (2)

2 row(s)

Diagnostic Type	Invocation Time	Result	Pages	Advisories	Last Updated	Details
OnDeviceAdfDiagnostics	04-Jun-2025 06:12:41	Success	2	0	04-Jun-2025 06:13:47	
OnDeviceAdfDiagnostics	04-Jun-2025 01:27:52	Success	2	1	04-Jun-2025 01:29:46	

Close

Status

HP Customer ID	513771
HP Connector ID	✓ 1463405
HP Device ID	156252241
Model	HP COLOR LASERJET MFP E78625
SKU	5QJ90A
IP Address	192.168.8.167
Hostname	wpp.localhost.12350.AQAAAAGOmGrgAgAADa3LOfL1
Last Update from HP	09-Jul-2025 15:07:50
Last Contact	✓ 09-Jul-2025 10:12:11
Ping Last Contact	✓ 09-Jul-2025 15:27:48

Actions

- View Model SDS Capabilities
- Refresh HP Data Cache
- Restart Device
- Update Firmware
- Configure Device
- Configure Device Message
- Remote Device EWS Access
- Configure Report A Problem App
- Print Quality Diagnostics
- View Advanced Support Information





HP SMART DEVICE SERVICES MANAGER - HP PRINT QUALITY DIAGNOSTICS

The screenshot shows the HP Smart Device Services Manager interface. A modal window titled "Device MXBCQ7B081: Print Quality Diagnostics" is open, displaying the "Audit Log Details" for the HP PQD Audit Log Id 1C1E3B50BF6243F69A53B9430EC896FB. The details include Diagnostic Type (OnDeviceAdfDiagnostics), Invocation Time (4 Jun 2024, 18:08:47), Result (Success), Pages (2), Engineering Advisories (PQTriageLightPrintYellow), and Last Updated (4 Jun 2024, 18:09:22). The "Printed Images" section is highlighted, and a "Download" link is visible. A callout box points to the "Download" link, stating "Download and view scanned PQD images". Another callout box points to the "Printed Images" section, stating "Download and view scanned PQD images".

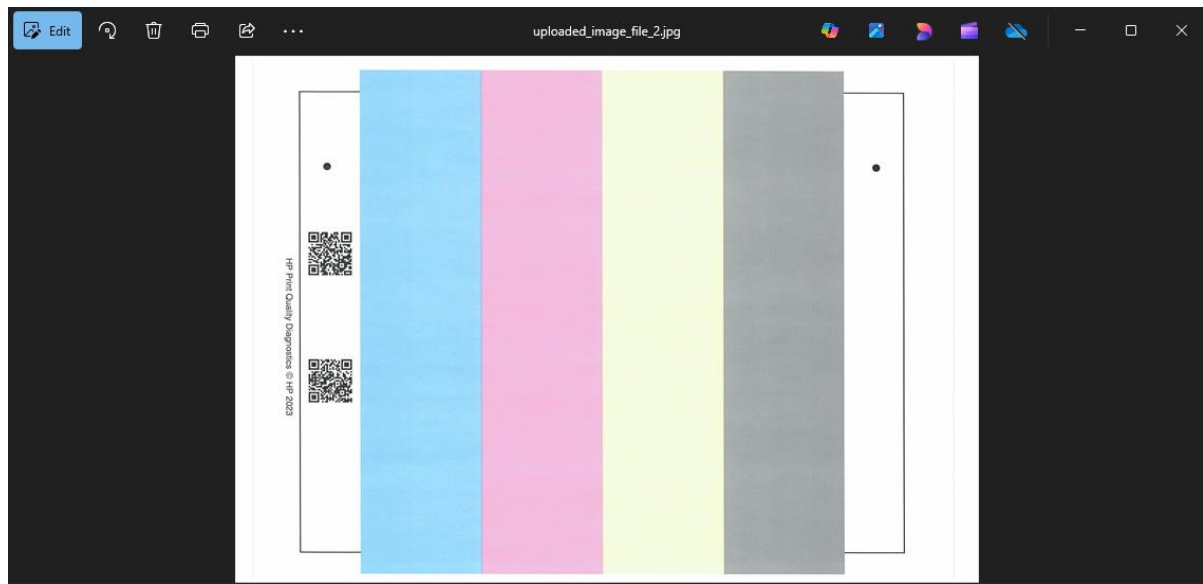
Download and view scanned PQD images

The screenshot shows the Windows File Explorer interface. The "Downloads" folder is selected, and a file named "PQD_155215621_1C1E3B50BF6243F69A53B9430EC896FB.zip" is listed. A callout box points to the file, stating "Extract the file".

Extract the file

The screenshot shows the Windows File Explorer interface. The "Downloads" folder is selected, and a subfolder named "PQD_155215621_1C1E3B50BF6243F69A53B9430EC896FB" is open. Inside the folder, two files are listed: "uploaded_image_file_1" (JPG File, 364 KB) and "uploaded_image_file_2" (JPG File, 370 KB). A callout box points to the files, stating "Open the JPG".

Open the JPG



* Example of HP Print Quality Diagnostic test page after downloading

Print Quality Diagnostics Engineering Advisories

During a Print Quality Diagnostic run, if any problems are detected then an Alert/Engineering Advisory with a new Type named '**PrintQualityDiagnostics**' will be generated. These engineering advisories will be processed in the same way that SDS Manager processes existing engineering advisory types.

Click the icon  under the 'Details' column to view the notification details:

Device CZBBRGX0N8: Engineering Advisories						
Mar 2025 (1) 1 row(s)						
Current State	Type	Code	Severity	Created	Last Updated	More Info
Open ⓘ	PrintQualityDiagnostics	PQTriageStreaksPrintPage	Medium	18 Mar 2025, 08:42:19	18 Mar 2025, 08:42:19	Triage streaks in all areas of printed documents including in unprinted areas 
Close						



Device CZBBR6X0N8: Engineering Advisories

Notification Details

HP Action Id: a79e519a-b2a7-4681-8cd0-71a3089130f7

Current State:

Type: PrintQualityDiagnostics

Code: PQTriageStreaksPrintPage

Severity: Medium

Created: 18 Mar 2025, 08:42:19

Last Updated: 18 Mar 2025, 08:42:19

Description: The printer is experiencing print quality issues due to streaking in all colors.

More Info: Triage streaks in all areas of printed documents including in unprinted areas

Parts:

- 527G4MC; HP LaserJet Mngd 110V Enhanced Fuser Kit
- 527G5MC; HP LaserJet Mngd 220V Enhanced Fuser Kit
- 527H0MC; HP LaserJet Managed Image Transfer Belt

Firmware Version: 2508385_053970

Print Quality Diagnostics

Diagnostic Type	Invocation Time
RemotePQDiagnostics	18 Mar 2025, 08:42:19

Status History

Date	State	User	Comment
18 Mar 2025, 08:42:19	Open	[HP]	

Add Comment

This is the link to the Audit Log entry for the diagnostic run that generated the advisory

Functionality Overview

	Inline Scan Bar	Inline Scan Bar Accessory	App
Remote Invocation	manual	manual	n/a
Firmware Invocation	automatic	n/a	n/a
Local Console Invocation	n/a	n/a	manual

	User Invocation	Automatic Firmware Invocation
Test Pages Printed	Yes	No
Test Pages Scanned	Yes	No
Test Pages Uploaded	Yes	No
Audit Logs Available	Yes	No



Image Scans Available	Yes	No
Engineering Advisories Generated	Yes (if faults found)	Yes (if faults found)

HP Print Quality Diagnostics Best Practices and Recommendations

The following best practices are recommended by HP for using HP Print Quality Diagnostics and will help you take full advantage of the capabilities of the solution.

Timing of Diagnostic Runs - Inline Scan Bar Remote Analysis (Hardware)

In general, print quality issues progress very gradually. In many cases, an issue may not be visible to the human eye for days or weeks depending on usage. As a result, it is generally not required to run [HP Print Quality Diagnostics - Inline Scan Bar Remote Analysis](#) operations more than once a week unless the device is very high-use or has specialised print quality use cases. To avoid interrupting end-user print jobs, it is recommended that diagnostic runs be performed during low-use time periods.

Out-of-Box Print Quality

Vibration, humidity, and high temperatures during shipping may impact a device's initial print quality. These print quality issues are generally represented by mild banding and decrease over time and pages printed. In general, print quality issues caused by environmental factors during shipping will resolve themselves within the first 300-400 pages of use.

If your device experiences initial print quality issues during its initial out-of-box life, HP recommends not replacing the cartridge, and monitoring the issue for the first few hundred pages of use to ensure the issue subsides as expected.

NOTE: Cartridges that do not ship inside the device do not experience these initial print quality issues.

Closing Engineering Advisories

Engineering Advisories can be closed with notes indicating whether the Action and/or documentation provided was accurate or helpful. By closing actions with this information, you will be helping HP monitor the accuracy of analysis models as well as the efficacy of the provided documentation. This will allow HP to continually improve their system and provide greater value to partners.



NOTE: There is a two-day gap between the closure of an Engineering Advisory, and the creation of a new Engineering Advisory, should the same fault recur, please see the following advice about checking faults are fixed before closing advisories.

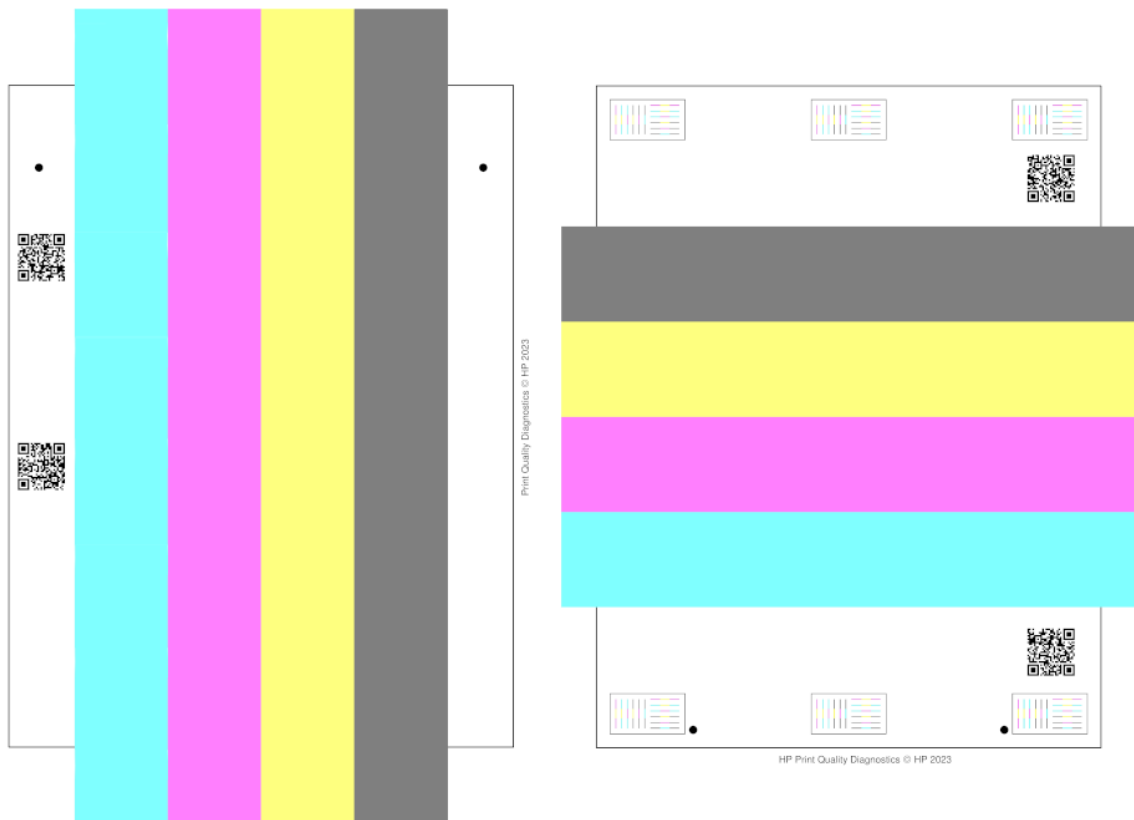
Recommended Checks Before Closing Engineering Advisories

Closed engineering advisories of type “PrintQualityDiagnostics” cannot be re-opened for two days after the advisory has been closed - even if the trigger condition is still present. As a result, the following order of operations are considered best practices for PQD engineering advisory resolution:

1. Service technician performs step(s) to resolve the issue.
2. Service technician performs another HP Print Quality Diagnostics run (using Inline Scan Bar or the HP Print Quality Diagnostics App)
3. Service technician inspects the results of the operation by viewing the Print Quality Diagnostics Audit Log entry.
 - If the Operation Audit Log indicates no engineering advisories were generated for that diagnostic run:
 - The service technician can close the engineering advisory.
 - If the new Print Quality Diagnostics Audit Log entry still contains a link to an engineering advisory:
 - The service technician returns to step 1.

End-User Communication

HP Print Quality Diagnostics - Inline Scan Bar – Remote Analysis (Hardware) uses diagnostic print jobs that will be visible by a device's end-users. These diagnostic pages will look like the following images:



It's important for end-users to understand that these pages are "normal" and are not anything to be concerned about. These pages can be recycled along with other paper items.

Printed Pages

Both HP Print Quality Diagnostics - Inline Scan Bar Remote analysis (Hardware) and the HP Print Quality Diagnostics App (Software) report the number of pages that were printed during the diagnostics run in the Operation Audit Log. This allows for full transparency on how many diagnostic pages were printed as well as the time of each diagnostic run. The Operation Audit Log also contains triage information if the operation failed as well as a listing of any actions that were detected based on the analysis of the images. Lastly, it contains a link allowing for the diagnostic run's images to be obtained.

Paper

- HP Print Quality Diagnostics requires A4 or Letter paper to be installed on the device under test. This is true even for A3-class devices.
- For A3-class devices using the Inline Scan bar, the A4 or Letter paper must be oriented in the tray, so the media feeds long-edge through the paper path.
- The A4 or Letter paper must be white.



- Printing diagnostic pages on letterhead paper is strongly discouraged.
- HP does not recommend recycled paper for use cases where print quality is of high importance.

Cloud DCA

HP recommends the use of a Cloud DCA to simplify authentication and for ensuring the data collector is always up to date with the latest revision.

Triage

The use of the Operation Audit Log is encouraged for triaging issues with HP Print Quality Diagnostics. This log provides useful information for identifying the cause of analysis run failures.